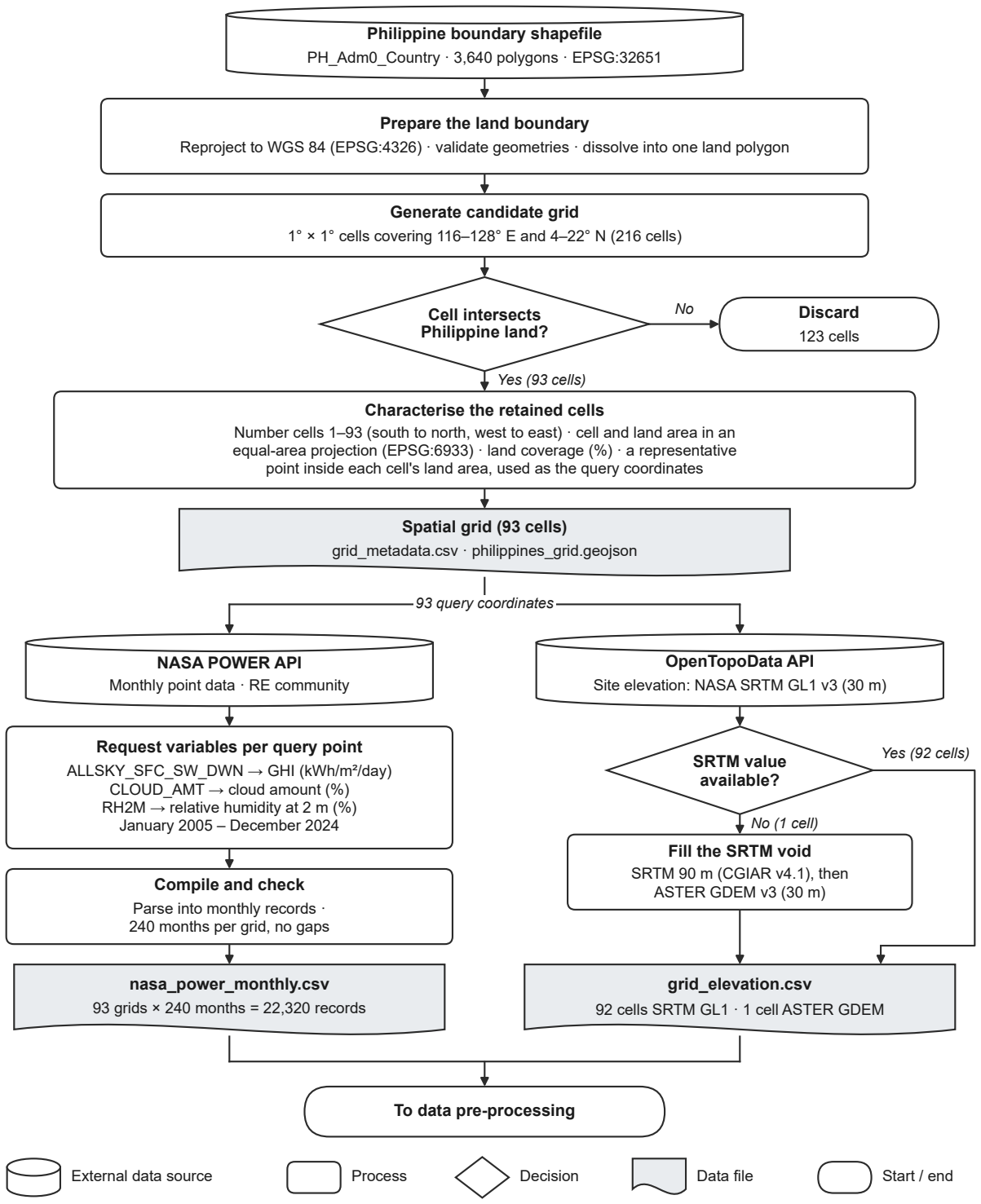

Philippine boundary shapefile
PH_Adm0_Country · 3,640 polygons · EPSG:32651
Prepare the land boundary
Reproject to WGS 84 (EPSG:4326) · validate geometries · dissolve into one land polygon
Generate candidate grid
1° × 1° cells covering 116–128° E and 4–22° N (216 cells)
Cell intersects Philippine land?
No
Discard
123 cells
Yes (93 cells)
Characterise the retained cells
Number cells 1–93 (south to north, west to east) · cell and land area in an equal-area projection (EPSG:6933) · land coverage (%) · a representative point inside each cell's land area, used as the query coordinates
Spatial grid (93 cells)
grid_metadata.csv · philippines_grid.geojson
93 query coordinates
NASA POWER API
Monthly point data · RE community
OpenTopoData API
Site elevation: NASA SRTM GL1 v3 (30 m)
Request variables per query point
ALLSKY_SFC_SW_DWN → GHI (kWh/m²/day)
CLOUD_AMT → cloud amount (%)
RH2M → relative humidity at 2 m (%)
January 2005 – December 2024
SRTM value available?
Yes (92 cells)
No (1 cell)
Fill the SRTM void
SRTM 90 m (CGIAR v4.1), then ASTER GDEM v3 (30 m)
Compile and check
Parse into monthly records · 240 months per grid, no gaps
nasa_power_monthly.csv
93 grids × 240 months = 22,320 records
grid_elevation.csv
92 cells SRTM GL1 · 1 cell ASTER GDEM
To data pre-processing
External data source
Process
Decision
Data file
Start / end